HACKERS DO BEM

FUNDAMENTAL - TURMA 20

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MÓDULO 12

ATIVIDADE PRÁTICA 2

Atividade 12.6: passo 8

Resultado do comando `exiftool contrato_08_2021.pdf` exibindo os metadados do documento PDF no Kali Linux.

```
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Image Size                      : 300×379
Megapixels                      : 0.114
Thumbnail Image                 : (Binary data 6473 bytes, use -b option to extract)

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# wget https://www.cnj.jus.br/wp-content/uploads/2021/07/contrato_08_2021.pdf
--2026-04-05 18:31:46--  https://www.cnj.jus.br/wp-content/uploads/2021/07/contrato_08_2021.pdf
Resolvendo www.cnj.jus.br (www.cnj.jus.br)... 13.226.209.106, 13.226.209.125, 13.226.209.65, ...
Conectando-se a www.cnj.jus.br (www.cnj.jus.br)|13.226.209.106|:443... conectado.
A requisição HTTP foi enviada, aguardando resposta... 200 OK
Tamanho: 22074478 (21M) [application/pdf]
Salvando em: "contrato_08_2021.pdf"

contrato_08_2021.pdf                100%[===================================================================================================>]  21,05M  16,3MB/s    em 1,3s

2026-04-05 18:31:48 (16,3 MB/s) - "contrato_08_2021.pdf" salvo [22074478/22074478]


┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# ls
contrato_08_2021.pdf  IPTCpanel.jpg

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# exiftool contrato_08_2021.pdf
ExifTool Version Number         : 13.36
File Name                       : contrato_08_2021.pdf
Directory                       : .
File Size                       : 22 MB
File Modification Date/Time     : 2022:03:07 15:23:03-03:00
File Access Date/Time           : 2026:04:05 18:31:48-03:00
File Inode Change Date/Time     : 2026:04:05 18:31:48-03:00
File Permissions                : -rw-r--r--
File Type                       : PDF
File Type Extension             : pdf
MIME Type                       : application/pdf
Linearized                      : No
Page Count                      : 626
Warning                         : Duplicate 'DR' entry in dictionary (ignored)
Has XFA                         : No
Language                        : pt-BR
Tagged PDF                      : Yes
XMP Toolkit                     : 3-Heights(TM) XMP Library 4.8.25.2 (http://www.pdf-tools.com)
Creator Tool                    : Microsoft® Word para Microsoft 365
Create Date                     : 2021:07:06 17:28:51-03:00
Modify Date                     : 2021:07:06 21:02:09Z
Metadata Date                   : 2021:07:06 21:02:09Z
Document ID                     : uuid:9EF429E5-0920-4C12-B890-66E34903C95E
Instance ID                     : urn:uuid:C816A848-8634-4059-B293-E647B1DFC85F
Producer                        : 3-Heights(TM) PDF Security Shell 4.8.25.2 (http://www.pdf-tools.com)
PDF Version                     : 1.7
Format                          : application/pdf

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─#
```

Atividade 12.7: passo 5

Resultado do comando `exiftool IPTCpanel.jpg` mostrando a remoção dos metadados da imagem após uso da opção `-all=`.

```
Color Transform                 : YCbCr
Image Width                     : 300
Image Height                    : 379
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Image Size                      : 300×379
Megapixels                      : 0.114

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# exiftool -AllDates="1917:12:12 06:00:00" "IPTCpanel.jpg"
    1 image files updated

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# exiftool IPTCpanel.jpg
ExifTool Version Number         : 13.36
File Name                       : IPTCpanel.jpg
Directory                       : .
File Size                       : 55 kB
File Modification Date/Time     : 2026:04:05 18:56:18-03:00
File Access Date/Time           : 2026:04:05 18:56:18-03:00
File Inode Change Date/Time     : 2026:04:05 18:56:18-03:00
File Permissions                : -rw-r--r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
Exif Byte Order                 : Big-endian (Motorola, MM)
Modify Date                     : 1917:12:12 06:00:00
Y Cb Cr Positioning             : Centered
Exif Version                    : 0232
Date/Time Original              : 1917:12:12 06:00:00
Create Date                     : 1917:12:12 06:00:00
Components Configuration        : Y, Cb, Cr, -
Color Space                     : Uncalibrated
Current IPTC Digest             : 86e482c077aedfe38855ec2781d06810
Copyright Notice                : Direitos autorais do Professor
Application Record Version      : 4
XMP Toolkit                     : Image::ExifTool 13.36
Creator                         : Prof. Max
Rights                          : Direitos autorais do Professor
DCT Encode Version              : 100
APP14 Flags 0                   : [14]
APP14 Flags 1                   : (none)
Color Transform                 : YCbCr
Image Width                     : 300
Image Height                    : 379
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Image Size                      : 300×379
Megapixels                      : 0.114

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─#
```

Atividade 12.8: passo 7

Resultado do comando `exiftool IPTCpanel.jpg` exibindo os novos metadados com autor e data modificados após utilização do parâmetro `-AllDates`.

```
Color Transform                 : YCbCr
Image Width                     : 300
Image Height                    : 379
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Image Size                      : 300×379
Megapixels                      : 0.114

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# exiftool -AllDates="1917:12:12 06:00:00" "IPTCpanel.jpg"
    1 image files updated

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# exiftool IPTCpanel.jpg
ExifTool Version Number         : 13.36
File Name                       : IPTCpanel.jpg
Directory                       : .
File Size                       : 55 kB
File Modification Date/Time     : 2026:04:05 18:56:18-03:00
File Access Date/Time           : 2026:04:05 18:56:18-03:00
File Inode Change Date/Time     : 2026:04:05 18:56:18-03:00
File Permissions                : -rw-r--r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
Exif Byte Order                 : Big-endian (Motorola, MM)
Modify Date                     : 1917:12:12 06:00:00
Y Cb Cr Positioning             : Centered
Exif Version                    : 0232
Date/Time Original              : 1917:12:12 06:00:00
Create Date                     : 1917:12:12 06:00:00
Components Configuration        : Y, Cb, Cr, -
Color Space                     : Uncalibrated
Current IPTC Digest             : 86e482c077aedfe38855ec2781d06810
Copyright Notice                : Direitos autorais do Professor
Application Record Version      : 4
XMP Toolkit                     : Image::ExifTool 13.36
Creator                         : Prof. Max
Rights                          : Direitos autorais do Professor
DCT Encode Version              : 100
APP14 Flags 0                   : [14]
APP14 Flags 1                   : (none)
Color Transform                 : YCbCr
Image Width                     : 300
Image Height                    : 379
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Image Size                      : 300×379
Megapixels                      : 0.114

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─#
```

Atividade 12.9: passo 5

Saída do comando `ils /dev/nvme0n1p1 2` mostrando as informações forenses do inode da partição analisada com The Sleuth Kit.

```
Partition table entries are not in disk order.


Disk /dev/nvme1n1: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: Amazon Elastic Block Store
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
Disklabel type: dos
Disk identifier: 0x9f2d034a

Device         Boot  Start    End Sectors  Size Id Type
/dev/nvme1n1p1         2048 206847  204800  100M 83 Linux
/dev/nvme1n1p2       206848 411647  204800  100M 83 Linux
/dev/nvme1n1p3       411648 821247  409600  200M 83 Linux

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# fls /dev/nvme0n1p1fls /dev/nvme0n1p1
Error stat(ing) image file (raw_open: image "/dev/nvme0n1p1fls" - Arquivo ou diretório inexistente)

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# fls /dev/nvme0n1p1
d/d 524461:     home
d/d 11: lost+found
d/d 131073:     boot
d/d 396737:     curso
d/d 524289:     var
d/d 393229:     etc
d/d 131075:     usr
l/l 13: bin
d/d 524460:     dev
l/l 14: lib
d/d 524463:     proc
d/d 524464:     root
d/d 524465:     run
l/l 16: sbin
d/d 393258:     sys
d/d 524467:     tmp
l/l 15: lib64
d/d 524556:     mnt
d/d 524557:     srv
d/d 524558:     opt
d/d 524560:     media
r/r 17: .autorelabel
d/d 410038:     snap
V/V 1433601:    $OrphanFiles

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# ils /dev/nvme0n1p1 2
class|host|device|start_time
ils|kali||1775424932
st_ino|st_alloc|st_uid|st_gid|st_mtime|st_atime|st_ctime|st_crtime|st_mode|st_nlink|st_size

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─#
```

Atividade 12.10: passo 5

Print da atividade 12.10: Resultado do comando `stat Backup_nvme1n1p3` exibindo os detalhes do arquivo criado pela cópia bit a bit da partição usando o comando dd.

```
└─#  fdisk -l
Disk /dev/nvme0n1: 22 GiB, 23622320128 bytes, 46137344 sectors
Disk model: Amazon Elastic Block Store
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
Disklabel type: gpt
Disk identifier: 9FFA977A-4F2F-44A0-BCAC-15A7352798A

Device           Start      End  Sectors  Size Type
/dev/nvme0n1p1  262144 46137310 45875167 21,9G Linux filesystem
/dev/nvme0n1p14   2048     8191     6144    3M BIOS boot
/dev/nvme0n1p15   8192   262143   253952  124M EFI System

Partition table entries are not in disk order.


Disk /dev/nvme1n1: 1 GiB, 1073741824 bytes, 2097152 sectors
Disk model: Amazon Elastic Block Store
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
Disklabel type: dos
Disk identifier: 0x9f2d034a

Device         Boot  Start    End Sectors  Size Id Type
/dev/nvme1n1p1        2048 206847  204800  100M 83 Linux
/dev/nvme1n1p2      206848 411647  204800  100M 83 Linux
/dev/nvme1n1p3      411648 821247  409600  200M 83 Linux

┌──(root㉿kali)-[~]
└─# dd if=/dev/nvme1n1p3 of=/home/aluno/Documentos/Arquivos/Backup_nvme1n1p3
409600+0 registos entrados
409600+0 registos saidos
209715200 bytes (210 MB, 200 MiB) copiados, 1,55615 s, 135 MB/s

┌──(root㉿kali)-[~]
└─# cd /home/aluno/Documentos/Arquivos

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# ls
Backup_nvme1n1p3

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─# stat Backup_nvme1n1p3
    Arquivo: Backup_nvme1n1p3
    Tamanho: 209715200  Blocos: 409600     bloco de E/S: 4096   regular file
Dispositivo: 259,5      Inode: 546462     Ligações: 1
     Acesso: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
     Acesso: 2026-04-05 18:37:28.198445805 -0300
Modificação: 2026-04-05 18:37:29.746439363 -0300
  Alteração: 2026-04-05 18:37:29.746439363 -0300
    Criação: 2026-04-05 18:37:28.198445805 -0300

┌──(root㉿kali)-[/home/aluno/Documentos/Arquivos]
└─#
```